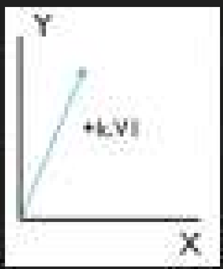
☆ Vectors

Data in tabular form ⇒ rows & columns

☆ Operations

① Addition (Dot Product)

② Scalar Multiplication

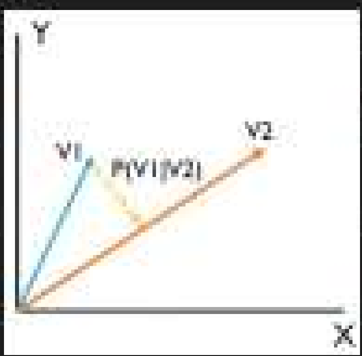


Positive K
Grows

Negative K
shrinks

③ Projection

Shadow a vector places on another vector.



⇒ We get to know unknown features V2 has more than V1

★ Matrices

① Matrix form from equations

Suppose we have 2 equations:

$2x + 2y = 10$

$4x + y = 18$

This in Matrix Form is:

$$\begin{bmatrix} 2 & 2 \\ 4 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 10 \\ 18 \end{bmatrix}$$

② Addition of Matrices

③ Subtraction of Matrices

④ Multiplication of Matrices

⑤ Transpose → to change dimensionality

⑥ Determinant → Product of Eigen Values

& scalar value of Matrix

⑦ Inverse of Matrix

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\quad A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ → 2x2

$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ $\quad A^{-1} = \frac{1}{\det(A)}\begin{bmatrix} \begin{vmatrix} e & f \\ h & i \end{vmatrix} & \begin{vmatrix} d & f \\ g & i \end{vmatrix} & \begin{vmatrix} d & e \\ g & h \end{vmatrix} \\ \begin{vmatrix} b & c \\ h & i \end{vmatrix} & \begin{vmatrix} a & c \\ g & i \end{vmatrix} & \begin{vmatrix} a & b \\ g & h \end{vmatrix} \\ \begin{vmatrix} b & c \\ e & f \end{vmatrix} & \begin{vmatrix} a & c \\ d & f \end{vmatrix} & \begin{vmatrix} a & b \\ d & e \end{vmatrix} \end{bmatrix}$ → 3x3

⑧ Vectors as Matrix

$v2 = 3x + 4y$

$[3 \;\; 4]\,[x \;\; y] = v$

⑨ VaM operations

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}\begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix}$$

Scaling

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}\begin{bmatrix} 1 & 0 \\ m & 1 \end{bmatrix}$$

Shearing

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}\begin{bmatrix} \cos a & -\sin a \\ \sin a & \cos a \end{bmatrix}$$

Rotation

⑩ Row Echlon Method (Row manipulation)

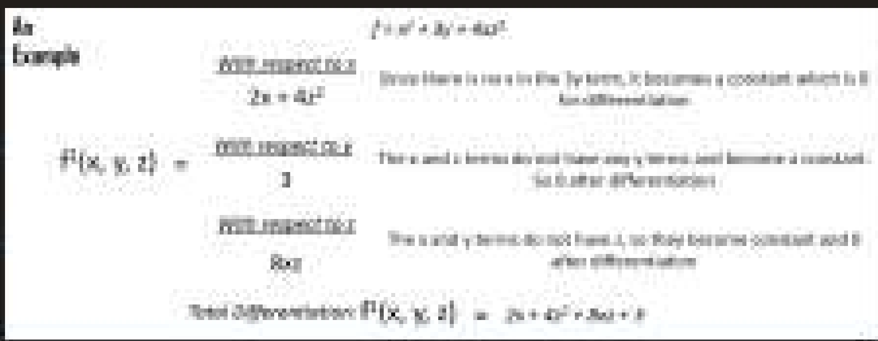
☆ Eigen Vectors

↳ do not change directions even if transformation applied to them.

☆ Multivariate Calculus

① Differentiation

- Power Rule
- Sum Rule
- Product Rule
- Partial Differentiation



② Jacobian

↳ helps find Global Maxima of datasets

③ Gradient Descent (notes)